

MATCH YOUR NOTES

Same formulas, simpler symbols

Source: Lecture 2, PDF pages 9 and 22-25 (title slide counts as page 1).

The notes use theta for the parameters and h for the prediction.

$$h_{\theta}(x) = g(\theta^T x) = p, \quad g(z) = \sigma(z) = \frac{1}{1 + e^{-z}}$$

$$\theta_0 = b, \quad \theta_1 = w, \quad x_0 = 1$$

$$\theta^T(1, x) = \theta_0 \times 1 + \theta_1 \times x = b + wx$$

Cost in the notes is L here: one example's binary cross-entropy.

J is the mean loss in both. The notes write log; here ln means natural log.

The learning rate alpha and simultaneous-update rule are unchanged.

Our choices for a small worked example

The three data pairs, starting values, and learning rate are chosen for practice.

START HERE

What are we trying to learn?

A number x goes in. We estimate the probability that its label y is 1.

Our three training examples

Example 1

$$(x_1, y_1) = (1, 0)$$

Example 2

$$(x_2, y_2) = (2, 1)$$

Example 3

$$(x_3, y_3) = (3, 1)$$

x = one input value. y = the known label, either 0 or 1.

w = weight: how strongly x affects the score. b = bias: a shared offset.

p = the predicted probability of label 1. It is not the known label y .

One shared w and b serve all 3 examples; i identifies the example.

$$w = 0, \quad b = 0, \quad \alpha = 0.1, \quad m = 3$$

Zero is our starting guess; alpha controls step size; m counts examples.

Our plan

Predict each p , measure the loss, calculate gradients, then update w and b .

STEP 1 / FORWARD PASS

Turn each x into a probability

Use the same starting $w = 0$ and $b = 0$ for all three examples.

$$z_i = wx_i + b \quad \longrightarrow \quad p_i = \sigma(z_i) = \frac{1}{1 + e^{-z_i}}$$

Sigmoid maps any score to a probability between 0 and 1.

The symbol sigma names sigmoid: negative z gives $p < 0.5$; positive z gives $p > 0.5$.

e is approximately 2.718; e to the power 0 equals 1.

$$z_1 = 0 \times 1 + 0 = 0 \quad p_1 = \frac{1}{1 + e^0} = \frac{1}{2} = 0.5$$

$$z_2 = 0 \times 2 + 0 = 0 \quad p_2 = \frac{1}{1 + e^0} = \frac{1}{2} = 0.5$$

$$z_3 = 0 \times 3 + 0 = 0 \quad p_3 = \frac{1}{1 + e^0} = \frac{1}{2} = 0.5$$

What does 0.5 mean?

Before learning, the model assigns a 50% chance of label 1 to every point.

STEP 2 / LOSS

How wrong are these probabilities?

Binary cross-entropy (BCE) penalizes low probability for the correct label.

$$L_i = -[y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)]$$

$\ln$ means natural logarithm. Use $\ln$ on your calculator, not $\log$ base 10.

$$L_1 = -[0 \ln(0.5) + 1 \ln(1 - 0.5)] = -\ln(0.5) = 0.693147$$

$$L_2 = -[1 \ln(0.5) + 0 \ln(1 - 0.5)] = -\ln(0.5) = 0.693147$$

$$L_3 = -[1 \ln(0.5) + 0 \ln(1 - 0.5)] = -\ln(0.5) = 0.693147$$

$$J = \frac{L_1 + L_2 + L_3}{3} = \frac{0.693147 + 0.693147 + 0.693147}{3} = 0.693147$$

Two useful shortcuts

If $y = 1$, use $-\ln(p)$. If $y = 0$, use $-\ln(1-p)$. J is the mean of all losses.

Read a derivative as sensitivity

We want to know which small changes would make the loss smaller.

$$\frac{\partial L}{\partial z}$$

Read this as: how much does loss L change per small increase in score z ?

The curly symbol means a partial derivative: hold other inputs fixed.

score $z \rightarrow$ probability $p \rightarrow$ loss L

For example 1 ($y = 0$), at $p = 0.5$:

A small $+0.01$ change in z gives about $+0.25 \times 0.01 = +0.0025$ in p .

That gives about $+2 \times 0.0025 = +0.005$ in L : the loss increases.

The chain rule multiplies the two sensitivities: $2 \times 0.25 = 0.5$.

Next: where do 0.25 and 2 come from?

Use the supplied derivative rules on the next page. These changes are local estimates.

STEP 3 / WHY THE GRADIENT SIMPLIFIES

Where does p minus y come from?

These derivative rules are supplied; combine them using the chain rule.

$$\frac{\partial L}{\partial p} = -\frac{y}{p} + \frac{1-y}{1-p}, \quad \frac{\partial p}{\partial z} = p(1-p)$$

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial p} \frac{\partial p}{\partial z}$$

$$= \left(-\frac{y}{p} + \frac{1-y}{1-p}\right)p(1-p)$$

$$= -y(1-p) + (1-y)p = p - y$$

$$p = 0.5, y = 0: \quad \frac{\partial L}{\partial p} = 2, \quad \frac{\partial p}{\partial z} = 0.25 \quad \Rightarrow \quad 2(0.25) = 0.5$$

$$p = 0.5, y = 1: \quad \frac{\partial L}{\partial p} = -2, \quad \frac{\partial p}{\partial z} = 0.25 \quad \Rightarrow \quad -2(0.25) = -0.5$$

This shortcut combines sigmoid AND BCE

The sigmoid derivative is already included. Do not multiply by $p(1-p)$ again.

STEP 4 / PARAMETER GRADIENTS

How should w and b change?

J is a mean loss, so its gradient is the mean of the contributions.

$$z = wx + b \Rightarrow \frac{\partial z}{\partial w} = x, \quad \frac{\partial z}{\partial b} = 1$$

$$\frac{\partial L_i}{\partial w} = (p_i - y_i)x_i, \quad \frac{\partial L_i}{\partial b} = p_i - y_i$$

Example	Error: p - y	Weight contribution: error times input
1	0.5 - 0 = +0.5	(+0.5) x 1 = +0.5
2	0.5 - 1 = -0.5	(-0.5) x 2 = -1.0
3	0.5 - 1 = -0.5	(-0.5) x 3 = -1.5

$$\frac{\partial J}{\partial w} = \frac{0.5 - 1.0 - 1.5}{3} = -\frac{2}{3}$$

$$\frac{\partial J}{\partial b} = \frac{0.5 - 0.5 - 0.5}{3} = -\frac{1}{6}$$

The bias contribution is the error itself because its multiplier is 1.

Subtract learning rate times gradient

Both new values use gradients calculated at the OLD w and b .

$$\begin{aligned}w_{\text{new}} &= w - \alpha \frac{\partial J}{\partial w} \\&= 0 - 0.1 \left(-\frac{2}{3} \right) = \frac{1}{15} \approx 0.066667\end{aligned}$$

$$\begin{aligned}b_{\text{new}} &= b - \alpha \frac{\partial J}{\partial b} \\&= 0 - 0.1 \left(-\frac{1}{6} \right) = \frac{1}{60} \approx 0.016667\end{aligned}$$

A negative gradient means a small increase in that parameter lowers mean loss. Subtracting moves in that direction; alpha controls how far we move.

One batch, one update

We used all 3 points before updating. Do not update after each row here.

STEP 6 / CHECK THE RESULT

Predict again with the new parameters

Keep exact fractions internally; the displayed results are rounded.

$$z_i = \frac{1}{15}x_i + \frac{1}{60}, \quad p_i = \frac{1}{1 + e^{-z_i}}$$

$$z_1 = \frac{1}{15} + \frac{1}{60} \approx 0.083333 \Rightarrow p_1 = \frac{1}{1 + e^{-0.083333}} \approx 0.520821$$

$$L_1 = -\ln(1 - 0.520821) \approx 0.735682 \quad x = 1, y = 0$$

$$z_2 = \frac{2}{15} + \frac{1}{60} \approx 0.150000 \Rightarrow p_2 = \frac{1}{1 + e^{-0.150000}} \approx 0.537430$$

$$L_2 = -\ln(0.537430) \approx 0.620957 \quad x = 2, y = 1$$

$$z_3 = \frac{3}{15} + \frac{1}{60} \approx 0.216667 \Rightarrow p_3 = \frac{1}{1 + e^{-0.216667}} \approx 0.553956$$

$$L_3 = -\ln(0.553956) \approx 0.590670 \quad x = 3, y = 1$$

$$J_{\text{new}} = \frac{0.735682 + 0.620957 + 0.590670}{3} \approx 0.649103$$

The mean loss fell from 0.693147. Lower mean BCE is the training objective.

WHAT TO REMEMBER

One training step, in plain language

Use this order whenever you solve a one-feature logistic-regression question.

- 1. Score** Multiply input by weight; add bias: $z = wx + b$.
- 2. Probability** Apply sigmoid: $p = 1 / (1 + \exp(-z))$.
- 3. Loss** Use the true label: $-\ln(p)$ for 1; $-\ln(1-p)$ for 0.
- 4. Contributions** Calculate $p - y$; multiply by x for the weight gradient.
- 5. Average** Add the 3 contributions for each parameter; divide each sum by 3.
- 6. Update** Subtract learning rate times each mean gradient, simultaneously.

The first point's loss went UP, but the average loss went DOWN.

One update need not improve every example or change classification accuracy.

For another step, start from the NEW w and b and recompute every probability.

Convention: predict label 1 when $p \geq 0.5$. Here accuracy stays 2/3.

TRY IT YOURSELF

A fresh three-point problem

Cover the next two pages and calculate one full-batch update.

$$(x_1, y_1) = (0, 0), \quad (x_2, y_2) = (1, 0), \quad (x_3, y_3) = (2, 1)$$

$$w = 0, \quad b = 0, \quad \alpha = 0.1, \quad m = 3$$

1. Calculate every score, probability, and binary cross-entropy loss.
2. Find each $p - y$ and each $(p - y)x$ contribution.
3. Average the contributions separately for w and b .
4. Update both parameters using the old gradients.
5. Recompute all three probabilities and the mean loss.

Explain the signs

Do w and b move in the same direction? Explain using the two gradients.

Predictions, gradients, and update

At $w = b = 0$ every score is 0, so every probability is 0.5.

$$L_1 = L_2 = -\ln(1 - 0.5), \quad L_3 = -\ln(0.5), \quad J \approx 0.693147$$

$$p_1 - y_1 = 0.5 - 0 = +0.5$$

$$(p_1 - y_1)x_1 = (+0.5)(0) = +0.0$$

$$p_2 - y_2 = 0.5 - 0 = +0.5$$

$$(p_2 - y_2)x_2 = (+0.5)(1) = +0.5$$

$$p_3 - y_3 = 0.5 - 1 = -0.5$$

$$(p_3 - y_3)x_3 = (-0.5)(2) = -1.0$$

$$\frac{\partial J}{\partial w} = \frac{0 + 0.5 - 1}{3} = -\frac{1}{6}, \quad \frac{\partial J}{\partial b} = \frac{0.5 + 0.5 - 0.5}{3} = \frac{1}{6}$$

$$w_{new} = 0 - 0.1(-1/6) = 1/60$$

$$b_{new} = 0 - 0.1(1/6) = -1/60$$

Opposite signs, opposite moves

The negative weight gradient increases w ; the positive bias gradient decreases b .

Check the fresh model

Use $w = 1/60$ and $b = -1/60$. Recompute; do not reuse the old probabilities.

$$z_1 = (0 - 1)/60 \approx -0.016667, \quad p_1 = 1/(1 + e^{-z_1}) \approx 0.495833$$

$$y_1 = 0: \quad L_1 = -\ln(1 - 0.495833) \approx 0.684849$$

$$z_2 = (1 - 1)/60 \approx 0.000000, \quad p_2 = 1/(1 + e^{-z_2}) \approx 0.500000$$

$$y_2 = 0: \quad L_2 = -\ln(1 - 0.500000) \approx 0.693147$$

$$z_3 = (2 - 1)/60 \approx 0.016667, \quad p_3 = 1/(1 + e^{-z_3}) \approx 0.504167$$

$$y_3 = 1: \quad L_3 = -\ln(0.504167) \approx 0.684849$$

$$J_{\text{new}} = (0.684849 + 0.693147 + 0.684849)/3 \approx 0.687615$$

Common mistakes

Use probabilities in BCE; average once; update together; keep full precision until display.